

Multinomial Coefficients

Section 2.3

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



$$2 + 3 + 2 = 7$$

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Today's lecture

1. Ordered Allocations

2. Multinomial Identities

3. The Multinomial Theorem

4. Permuting Multisets

More than two classes

Question

How many ways can 10 distinct students be assigned to three labeled groups of sizes 4, 3, 3?

Choose the first group, then the second, then the third:

$$\binom{10}{4} \binom{6}{3} \binom{3}{3} = \frac{10!}{4!3!3!} = 4200.$$

Multinomial coefficients

Definition

If $n_1 + \dots + n_r = n$ with each $n_i \geq 0$, define

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1! n_2! \dots n_r!}.$$

This counts allocations of n distinct objects to r labeled boxes with specified occupancies $n_1, \dots, n_r$.

Choosing the boxes successively

$$\binom{n}{n_1} \binom{n - n_1}{n_2} \cdots \binom{n - n_1 - \cdots - n_{r-1}}{n_r} = \binom{n}{n_1, \dots, n_r}.$$

Factorials cancel in the product. Every allocation appears exactly once.

A permutation proof

Proof

Order all n objects, then cut the ordering into consecutive blocks of sizes $n_1, \dots, n_r$. There are $n!$ orderings.

Permuting objects within a block does not change the allocation. Each allocation arises from $n_1! \cdots n_r!$ orderings. Divide by this common multiplicity.

Labeled and unlabeled groups

Question

How many ways can six people be divided into three unlabeled pairs?

Labeled pairs give $6!/(2!)^3 = 90$ allocations. Each unlabeled pairing has $3!$ labelings, so the answer is

$$\frac{6!}{(2!)^3 3!} = 15.$$

Dividing by $3!$ is justified because all three groups have the same required size.

$$\binom{n}{n_1, n_2, \dots, n_r} = \binom{n}{n_{\sigma(1)}, n_{\sigma(2)}, \dots, n_{\sigma(r)}}$$

for every permutation σ of the box labels.

Relabeling the boxes is a bijection between the corresponding allocations.

A multinomial recurrence

Identity

If $n_1 + \cdots + n_r = n \geq 1$, then

$$\binom{n}{n_1, \dots, n_r} = \sum_{i: n_i > 0} \binom{n-1}{n_1, \dots, n_i - 1, \dots, n_r}.$$

Proof

Fix one object and distinguish the box containing it. Then allocate the other $n - 1$ objects.

The multinomial theorem

Theorem

For every nonnegative integer n ,

$$(x_1 + \cdots + x_r)^n = \sum_{n_1 + \cdots + n_r = n} \binom{n}{n_1, \dots, n_r} x_1^{n_1} \cdots x_r^{n_r},$$

where the sum is over nonnegative integers $n_1, \dots, n_r$.

Proof of the multinomial theorem

Proof

In the product of n factors $(x_1 + \cdots + x_r)$, assign each factor to the term chosen from it.

To obtain $x_1^{n_1} \cdots x_r^{n_r}$, assign n_i factors to x_i for each i . The number of assignments is the multinomial coefficient.

An expansion

$$\begin{aligned}(x + y + z)^3 &= x^3 + y^3 + z^3 \\ &\quad + 3x^2y + 3x^2z + 3y^2x + 3y^2z + 3z^2x + 3z^2y \\ &\quad + 6xyz.\end{aligned}$$

The coefficients add to $3^3 = 27$.

The coefficient of xyz is $\binom{3}{1,1,1} = 6$.

Question

Find the coefficient of x^2y^3z in $(2x - y + 3z)^6$.

Choose two $2x$ terms, three $-y$ terms, and one $3z$ term:

$$\binom{6}{2, 3, 1} 2^2 (-1)^3 3 = -720.$$

Summing multinomial coefficients

Identity

$$\sum_{n_1 + \dots + n_r = n} \binom{n}{n_1, \dots, n_r} = r^n.$$

Proof

Set every variable equal to 1 in the multinomial theorem. Alternatively, count all assignments of n distinct objects to r labeled boxes by their occupancies.

Words with prescribed multiplicities

Theorem

The number of words containing n_i copies of symbol i , for $i = 1, \dots, r$, is

$$\frac{(n_1 + \dots + n_r)!}{n_1! \dots n_r!}.$$

Choose which positions receive each symbol. Equal symbols are indistinguishable.

Rearranging MISSISSIPPI

Example

The word MISSISSIPPI has 11 letters: one M, four I's, four S's, and two P's.

Its distinct rearrangements number

$$\frac{11!}{1!4!4!2!} = 34,650.$$

Rearranging BOOKKEEPER

Example

BOOKKEEPER has 10 letters: two O's, two K's, three E's, and one each of B, P, R.

$$\frac{10!}{2!2!3!} = 151,200.$$

Counting the multiplicities correctly comes before applying the formula.

Multidimensional lattice paths

Question

How many paths from $(0, 0, 0)$ to (a, b, c) take positive unit steps along coordinate axes?

Encode a path by a word with a X's, b Y's, and c Z's. Therefore the answer is

$$\binom{a+b+c}{a, b, c}.$$

Using only some of a multiset

Question

How many three-letter words can be made from $\{a, a, a, b, b, c\}$ without exceeding the available multiplicities?

A single multinomial coefficient does not apply: different words can use different numbers of each letter.

Split into cases according to the multiplicity pattern.

Partial words: the cases

Pattern	Number	Explanation
Three equal	1	Only <i>aaa</i>
Two equal, one different	$4 \cdot 3 = 12$	<i>aab, aac, bba, bbc</i> , each reordered
All different	$3! = 6$	Rearrangements of <i>abc</i>

Total: $1 + 12 + 6 = 19$.

The general formula for partial words

Formula

From a multiset having m_i copies of symbol i , the number of words of length k is

$$\sum_{\substack{n_1 + \dots + n_r = k \\ 0 \leq n_i \leq m_i}} \frac{k!}{n_1! \cdots n_r!}.$$

First specify the multiplicities used; then count the words with those multiplicities.

A distinguished object in a box

Identity

When $n_i > 0$,

$$n_i \binom{n}{n_1, \dots, n_r} = n \binom{n-1}{n_1, \dots, n_i-1, \dots, n_r}.$$

Proof

Count an allocation together with a distinguished object in box i . Choose the allocation first, or choose the distinguished object first.

Question

How many words have four A's, three B's, and two C's, with the two C's adjacent?

Treat the adjacent C's as one block.

Proof

Replace CC by a single new symbol. We now arrange eight objects: four A's, three B's, and one CC block. Hence

$$\frac{8!}{4!3!} = 280.$$

Expanding the block is a bijection back to the required words.

- Multinomial coefficients count prescribed allocations into labeled boxes.
- They are the coefficients of powers of sums with several terms.
- The same formula counts words with repeated symbols.
- Partial words require a sum over the multiplicities actually used.

The Pigeonhole Principle